

5.55.2 Mechanical data

Technical data	Cube67+ DIO12 IOL4 E 8xM12 Art.-No. 56765
Ambient conditions	
Operating temperature	0 °C ... 55 °C
Storage temperature	-25 °C ... 70 °C
Flammability class	UL 94 V0
Materials	
Enclosure	PBT (UL94, V0), black
O-ring M12	Viton, green
Luminous comb+luminous ring	PC (UL94, V0)
Mechanical data	
Plug-in cycles / contact	≤50
Degree of protection EN 60629	IP67 (only if the device is plugged in and fastened by means of screws)
Vibration, sinusoidal EN 60068-2-6	10 ... 58 Hz; const. amplitude 0.35 mm
	58 ... 150 Hz; const. acceleration 15 g
Shock, semi-sinusoidal EN 60068-2-27	50 g / 11 ms
Processing notes	
Weight	210 g
Torques	
Round connector M12	0.6 +0.1 Nm
Fastening screw M4	2 ±0.5 Nm
Connection options	
System connection	M12 - connector (A-coded) 6-pole
I/O cables	8 x M12 - round connector (A-coded) 5-pole

5.55.3 Electrical data

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Max. number of inputs	16
Max. number of outputs	16
Operating voltage	24 V DC ±25 %
Current consumption	≤100 mA

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Sensor power supply			
Max. current	≤200 mA per M12 port		
Overload / short-circuit fuse for sensors	Multifuse (for each M12 port)	≤100 mA	Automatic restart
		>100 mA	Reset required
Overload / short-circuit fuse for IO link	Multifuse (for each M12 port)	≤300 mA	Automatic restart
		>300 mA	Reset required
	Multifuse trigger time 1 s at I _k ≥1 A and an ambient temperature of 23 °C		
Reverse polarity protection			
- Device electronics	Yes		
- Sensors	Yes		
- Actuators	Yes		
Outputs			
Rated current	≤1.6 A per channel		
Max. total current	≤4 A per supply		
Overvoltage protection	Yes (varistor)		
Cable length	0.75 mm²	≤10 m	
	0.34 mm²	≤5 m	
Conductor cross-section	≤1.5 mm²		
Signal delay	2 to 5 ms		
Max. switching frequency with resistive load	50 Hz		
Max. switching frequency with inductive load	5 Hz		
Max. lamp load	≤30 W		
Overload / short-circuit fuse	Electronic short-circuit detection tripping time <10 ms		
Inputs			
Input characteristic curve	IEC 1131-2, type 2		
Input filter	Approx. 1 ms		
Signal delay	2 ... 4 ms		
Overvoltage protection	Yes (varistor)		

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EMC		
	EN 61000-4-2 ESD	Contact ± 4 kV; air ± 8 kV
	EN 61000-4-3 HF field + GSM	10 V/m
	EN 61000-4-4 Burst	± 1 kV
	EN 61000-4-5 Surge	Asym./sym. ± 0.5 kV DC mains input
		Asym. ± 1 kV signal connections
	EN 61000-4-6 HF asymmetric	10 V
	EN 61000-4-8 Magnetic field 50 Hz	30 A/m
	EN 61000-6-3 Interference field strength	QP 30 dB μ V/m (30 ... 230 MHz) class B
		QP 37 dB μ V/m (230 ... 1000 MHz) class B
Rated voltage	IEC 60664-1	
Insulation resistance	IEC 60512-2	
Volume resistance	≤ 5 m Ω	